

NCP-AIN Spectrum CLI, Validation and Host-Side Ops Guide

Multi-page quick-reference for NVIDIA-Certified Professional AI Networking prep and day-to-day operations. Focus: Spectrum switch CLI on Onyx, RoCE validation, host-side InfiniBand/RDMA tests, and physical deployment checks.

Current exam scope from NVIDIA: AI Data Center Design and Optimization, NVIDIA Spectrum Networking, NVIDIA InfiniBand Networking, Kubernetes Integration, Troubleshooting Tools, Automation and Configuration.

Exam / Ops Focus Areas

Area	What to know	Priority
Management	Bring-up, image/version, mgmt0, SSH, save config, logs, inventory, environment.	High
Interfaces	Status, speed/FEC/MTU, counters, optics, LLDP, link validation.	High
L2/L3	VLAN, trunks, LACP, MLAG, SVI, routing basics, neighbor checks.	High
RoCE	Lossless Ethernet, PFC, ECN, jumbo MTU, queue behavior, congestion symptoms.	High
Host tools	ibstat, ibv_devinfo, ibdev2netdev, rdma link, ip/ethtool, ib_write_bw.	High
Automation	Jump-start, ZTP awareness, repeatable validation flow.	Medium

Recommended Onyx Workflow

```
switch > enable
switch # configure terminal
switch (config) # hostname sw-gpu-leaf01
switch (config) # interface mgmt0
switch (config interface mgmt0) # ip address 10.10.10.21/24
switch (config interface mgmt0) # no shutdown
switch (config interface mgmt0) # exit
switch (config) # ip default-gateway 10.10.10.1
switch (config) # copy running-config startup-config
```

Use **configuration jump-start** for guided initial setup. Validate management with **show interfaces mgmt0** before touching the data plane.

Spectrum / Onyx Switch Day-to-Day CLI

Core health, optics, cabling and data-plane commands.

High-Value Show Commands

Command Group	Command examples	Use
System	show version show inventory show system show environment show uptime	HW/SW
Management	show interfaces mgmt0 show users show logging show clock	OOB
Interfaces	show interfaces status show interfaces ethernet 1/1 show interfaces ethernet 1/1 counters	Link
Optics	show interfaces ethernet 1/1 transceiver show interfaces ethernet 1/1 diagnostics	DOM
Neighbors	show lldp show lldp neighbors show lldp neighbors interface ethernet 1/1	Patch
L2/L3	show vlan show mac-address-table show arp show ip route	Forwarding
Resiliency	show lacp show interfaces port-channel show mlag show mlag interfaces	HA
QoS/RoCE	show qos show dcb priority-flow-control show dcb ets show roce	Lossless

Representative Config Snippets

```
! Server-facing port
interface ethernet 1/1
description GPU-R01-N01-NIC1-RAIL-A
mtu 9216
speed 200G
no shutdown
```

```
! VLAN trunk example
vlan 100
name STORAGE
exit
interface ethernet 1/49
switchport mode trunk
switchport trunk allowed-vlan add 100,200,300
no shutdown
```

```
! Port-channel example
interface port-channel 10
description SPINE-UPLINK
exit
interface ethernet 1/31
channel-group 10 mode active
```

```
interface ethernet 1/32
 channel-group 10 mode active
```

```
! SVI example
interface vlan 100
 ip address 10.100.0.1/24
 no shutdown
```

Fast Validation Flow

```
show version
show interfaces status
show interfaces ethernet 1/1
show interfaces ethernet 1/1 counters
show interfaces ethernet 1/1 transceiver
show lldp neighbors interface ethernet 1/1
show logging
copy running-config startup-config
```

RoCE / AI Fabric Validation

Use this page for Spectrum-X / RoCE fabrics that carry east-west AI training traffic. Main failure domains are MTU mismatch, wrong priority/QoS mapping, missing PFC, congestion, FEC mismatch, or optics/cable faults.

RoCE Checks

Check	Command or validation	Goal
Jumbo MTU	show interfaces ethernet 1/1	9216 end-to-end
PFC	show dcb priority-flow-control	Lossless on RoCE priority
ETS/QoS	show dcb ets show qos	Queue mapping
RoCE state	show roce	Default profile / state
Counters	show interfaces ethernet 1/1 counters	No CRC/discards
LLDP peer	show lldp neighbors interface ethernet 1/1	Correct patching
Congestion	show logging + queue/counter review	Pause/ECN symptoms

What Good Looks Like

Ports are **admin up / oper up**, expected speed/FEC is negotiated, LLDP lands on the expected GPU NIC or peer switch, MTU is consistent, CRC/discard counters stay flat during burn-in, and PFC/ECN policy is applied consistently to the RoCE traffic class.

Troubleshooting Sequence for a Bad GPU-Rack Link

```
show interfaces ethernet 1/1
show interfaces ethernet 1/1 counters
show interfaces ethernet 1/1 transceiver
show lldp neighbors interface ethernet 1/1
show qos
show dcb priority-flow-control
show dcb ets
show roce
show logging
```

Interpretation: down/down usually means admin or physical issue; up/down often points to optics, cable, far-end, or speed/FEC mismatch; rising CRC/discards usually means dirty fiber, bad optic, bad DAC/AOC, or lane/FEC problems.

GPU Server / Host-Side Validation

These commands are useful on the AI server side for NIC, RDMA and fabric checks. They complement switch-side validation and are highly relevant for daily operations.

Basic Host Network Checks

```
hostname
ip addr
ip -br link
ip route
ethtool <netdev>
ethtool -S <netdev>
lspci | egrep -i 'mellanox|nvidia|connectx|bluefield'
nvidia-smi
nvidia-smi topo -m
```

RDMA / InfiniBand / RoCE Device Checks

```
ibstat
ibv_devinfo
ibdev2netdev
rdma link show
rdma dev show
rdma resource show
cat /sys/class/infiniband/*/ports/*/state
cat /sys/class/infiniband/*/ports/*/link_layer
```

Key things to verify: device is present, port state is **Active**, link layer is what you expect (**Ethernet** for RoCE or **InfiniBand** for IB), and netdev-to-HCA mapping matches the cutsheet.

Bandwidth / Latency Tests

Server side:

```
ib_write_bw -d mlx5_0 -i 1
```

Client side:

```
ib_write_bw -d mlx5_0 -i 1 <server-ip-or-lid>
```

Useful variants:

```
ib_write_bw -d mlx5_0 -i 1 -F --report_gbits <peer>
ib_read_bw -d mlx5_0 -i 1 -F --report_gbits <peer>
ib_send_bw -d mlx5_0 -i 1 -F --report_gbits <peer>
ib_write_lat -d mlx5_0 -i 1 <peer>
ib_read_lat -d mlx5_0 -i 1 <peer>
```

For RoCE, ensure the host can reach the peer IP, lossless policy is correct, and NIC affinity/NUMA placement are sane before trusting results.

GPU + Fabric Context Checks

```
nvidia-smi topo -m
numactl -H
lsmod | egrep 'mlx5|ib|rdma'
dmesg | egrep -i 'mlx5|rdma|ib|nvidia'
```

Deployment, Validation and Exam Notes

Rack Turn-Up Sequence

Step	Action	Output
1	Validate image, version, mgmt0, inventory, environment.	Switch healthy
2	Patch server NICs by rail and validate LLDP.	Correct peer map
3	Check optics/DAC/AOC and interface counters.	Clean physical layer
4	Validate MLAG/LACP or uplink health to spine.	Fabric reachable
5	Confirm MTU/PFC/ECN/RoCE settings.	Lossless ready
6	Run host-side ibstat / ibv_devinfo / ibdev2netdev.	Host sees NIC/HCA
7	Run ib_write_bw / latency tests.	Data path proven
8	Save config and capture acceptance evidence.	Operational handoff

High-Probability NCP-AIN Questions

Expect questions around rail redundancy, front-end vs back-end fabrics, RoCE vs InfiniBand use cases, lossless Ethernet design, interface and optics validation, MLAG/LAG basics, AI data center layout, and practical troubleshooting. For Spectrum specifically, be able to explain what you would check on the switch and what you would check on the GPU server.

Primary Sources Used

NVIDIA NCP-AIN certification page, NVIDIA certification overview, NVIDIA networking docs hub, NVIDIA Onyx User Manual, Onyx Getting Started, Onyx Ethernet Switching, Onyx RoCE, NVIDIA Cumulus Linux docs, and NVIDIA NVOS docs for InfiniBand switches.

URLs: <https://www.nvidia.com/en-us/learn/certification/ai-networking-professional/> | <https://www.nvidia.com/en-us/learn/certification/> | <https://networking-docs.nvidia.com/software/switch-software> | <https://networking-docs.nvidia.com/onyxum> | <https://networking-docs.nvidia.com/onyxum/3.10.4800/getting-started> | <https://networking-docs.nvidia.com/onyxum/3.10.4800/ethernet-switching> | <https://networking-docs.nvidia.com/onyxum/3.10.4800/rdma-over-converged-ethernet-roce> | <https://docs.nvidia.com/networking-ethernet-software/cumulus-linux/> | <https://networking-docs.nvidia.com/nvosinfinibandum>